INTERNET2 TECHEX26

OCT 26-30 | MINNEAPOLIS MN

Practical AI for Technical Professionals

Name of Presenter, Title of Presenter, Name of Organization

Date Here

Agenda

- Part I (Frank): Everything you need to know as a technical professional.
- Part II (Shannon): Putting it into practice with labs.

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Lab 1 - Installing and Configuring a CLI Harness



Selecting a Harness: Crush or Claude Code?

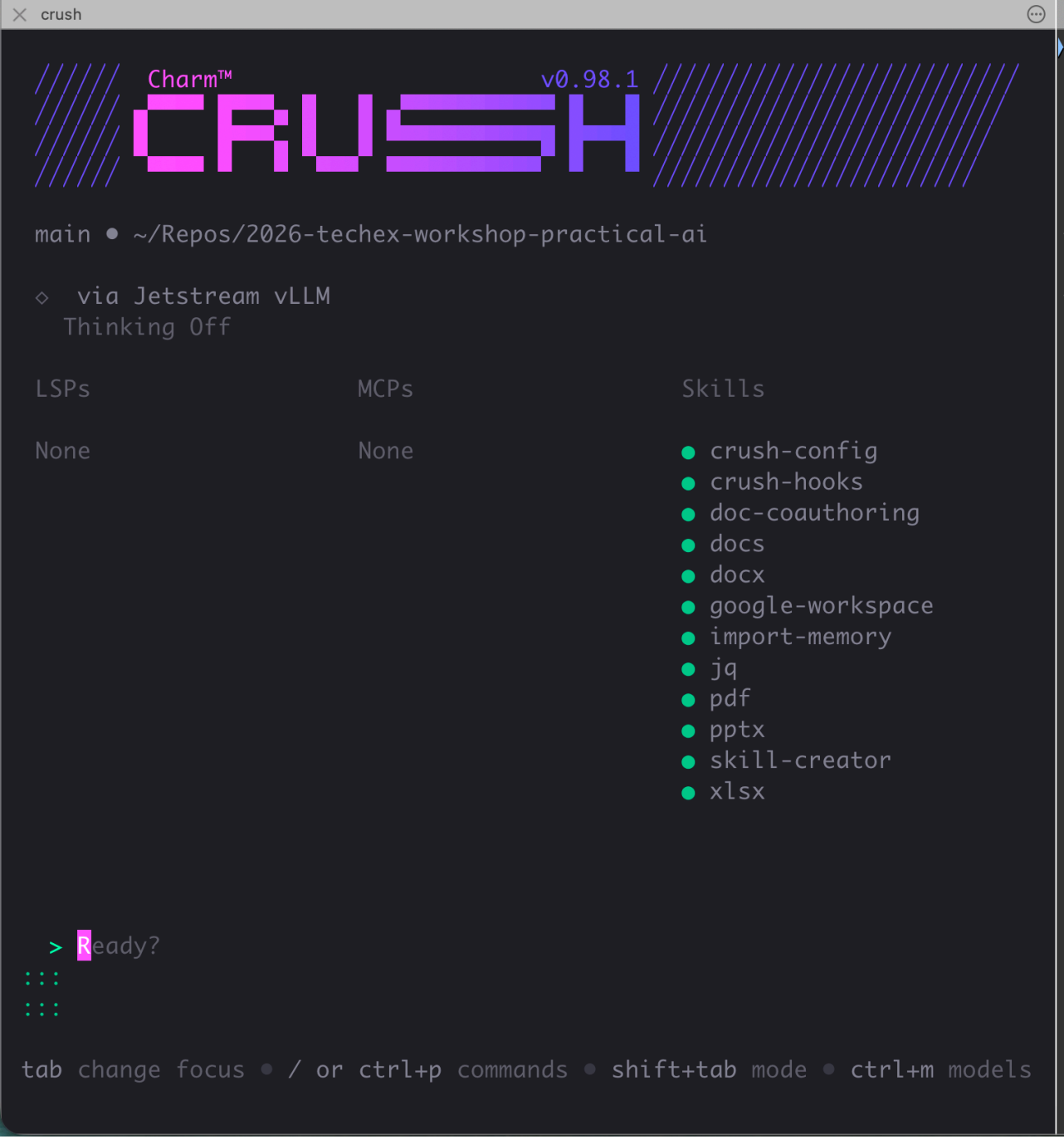
I'd like to use this workshop's LLM.	I brought my own Claude subscription.	I brought my own non-Claude subscription or API key
Follow hand-out for downloading **Crush**, which uses an API key.	Follow hand-out for downloading **Claude Code**, which uses OAuth.	Use your preferred terminal-based harness. If you are unsure, follow hand-out for downloading **Crush**
Open-source, works with many different models and model providers. Also works with OpenAI subscriptions.	Closed-source, is the only terminal-based option for Claude subscriptions.	Varies by harness. Most are open-source.

Installation and Configuration Instructions

Whether you are on Windows, Mac, or Linux, you can find instructions on installing Crush or Claude Code on the Workshop Repo on the front page (README.md) under “**Handouts**”

https://github.com/Internet2/2026-techex-workshop-practical-ai

Open the “**Handouts**” link to Google Doc.





Note: Yours won't look exactly like this!

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Things LLMs are Good For



Rubber Duck that Talks Back

- “Rubber Duck Debugging”
- More than just programming. It helps in many areas of work.
- Stuck on a hard problem? Writer’s block? Ask it questions.
- Ask it to interview you about a topic. Find gaps.
- Teacher with infinite patience. Ask it to explain something to you at the level you need.



Synthesis and Transformation

- Take fast, raw notes at a class or conference, and let an LLM clean it up, structure, and organize it. (Trip reports, anyone?)
- Ask the LLM to distill a conversation or “rubber duck session" into notes for future use.
- Ask the LLM to summarize logs and surface patterns.
- Ask the LLM to create visual representations based on notes.
- If you aren’t a note-taker now, LLMs might make you one.



Research Deep Dives



- LLMs can be very good at researching and finding academic papers or finding knowledge online.
- Can be instructed to highlight strong versus weak claims (with sources), verify peer review, locate critiques, consider bias or funding sources, and cross reference other academic papers.
- Schedule a routine or weekly deep dive to stay on top of the latest research for a topic you care about.

Coding (Had to do one slide on this…)

- LLMs are **non-deterministic**, but they can create **deterministic** code that a human can review.
- Can write predictable-looking code (for better or worse). This is continuing to improve as LLMs and their harnesses improve.
- Can be a *very* respectable code reviewer and catch things before you ask a human to review.
- Can explain code well and teach you implementation patterns.
- Fantastic at creating Proof of Concepts and exploration of ideas with little hand-holding.
  - Mature, production-grade coding is still not there yet without substantial steering and hand-holding. Also security considerations, expertise loss in humans, etc.

Building Small, Personal Scripts or Apps (Examples)

- Scheduled Topic Watcher
  - Could be a research topic where you want to stay up on the latest papers and want a summary each week.
  - Watch for board game sales at various retailers without having to subscribe to each website's newsletter.

- Create flashcards based on your notes for something new you are learning.

- If you take daily notes…
  - Synthesize a weekly "what did I do" for work.
  - Big read-later list? Summarize and organize the list so it is easier to prioritize which things to read first.

Building a Personal Knowledge Base

- Instruct an LLM to create a personal KB based on your notes and knowledge.
- You can also use research deep-dives to populate parts of your KB.
- The KB can also be used to make an LLM context-aware about your job and work, or to become “proficient” in some topic that is personal to you.
- This is Lab 1!



Golden Rules

1. Do not offload work you want to stay good at to an LLM.

2. You are accountable for what you produce using an LLM.

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Lab 2: Personal LLM Wiki



What is a Personal LLM Wiki?

- It can turn a “blank slate” model into one that knows the context of your work and/or life.
- It makes everything easier (and more fun!) when using LLMs for day-to-day work.
  - Less repeating of facts thanks to “memory”.

- This lab uses Andrej Karpathy’s methodology, which has been posted for LLM-ingestion on GitHub.

Under “**Tutorial Links**” on the GitHub, select “**Karpathy’s LLM Wiki Gist**”
(Leave it open; we will come back to it later)

What We Need

A friendly **Markdown** editor

An LLM **Harness**

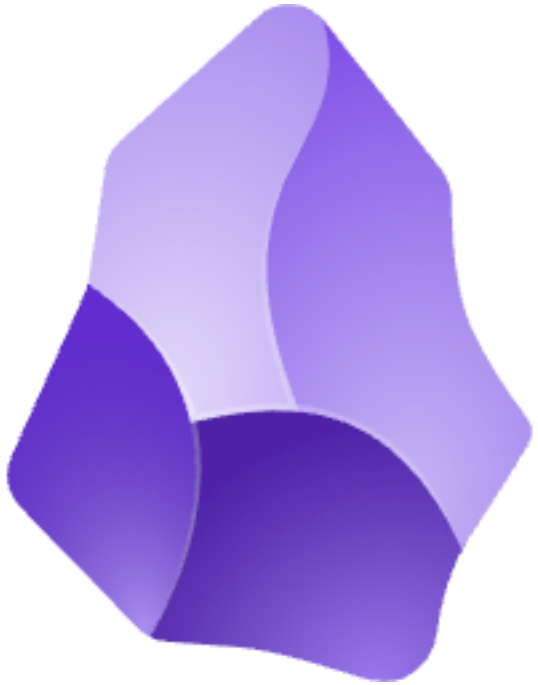
Access to a **Model**

An empty folder, or folder of stuff, for them to point to.

What is Obsidian?

- A friendly Markdown editor we will use in this lab.
- Existed before LLMs, and consequently markdown, were popularized by the public.

- https://obsidian.md/about



Clippings
Daily
Ideas
Writing is telepathy
Meta
Projects
References
Evergreen notes
Calmness is a superpo...
Travel
Creativity is combinato...
Emergence
Recipes
Books
Health
What if it were easy
Tools
Specialization is for ins...
First principles
Philosophy

Notes

Writing is telepathy | Evergreen notes

Graph of Writing is t...

Writing is telepathy

#evergreen

From On Writing

Ideas can travel through time and space

Ideas can travel through time and space without being uttered out loud. The process of telepathy requires two places:

- A **sending place**, a transmission place — where the writer sends ideas, such as a desk
- A **receiving place** — where the reader receives the ideas/imagery such as a couch, a comfortable chair, in bed

Quote

> Look, here's a table covered with red cloth. On it is a cage the size of a small fish aquarium. In the cage is a white rabbit with a pink nose and pink-rimmed eyes. On its back, clearly marked in blue ink, is the numeral 8. The most interesting thing here isn't even the carrot-munching rabbit in the cage, but the number on its back. I never opened my mouth and you never opened yours. We're



1 backlink 206 words

11:29

Japan Trip Planning

#projects #travel

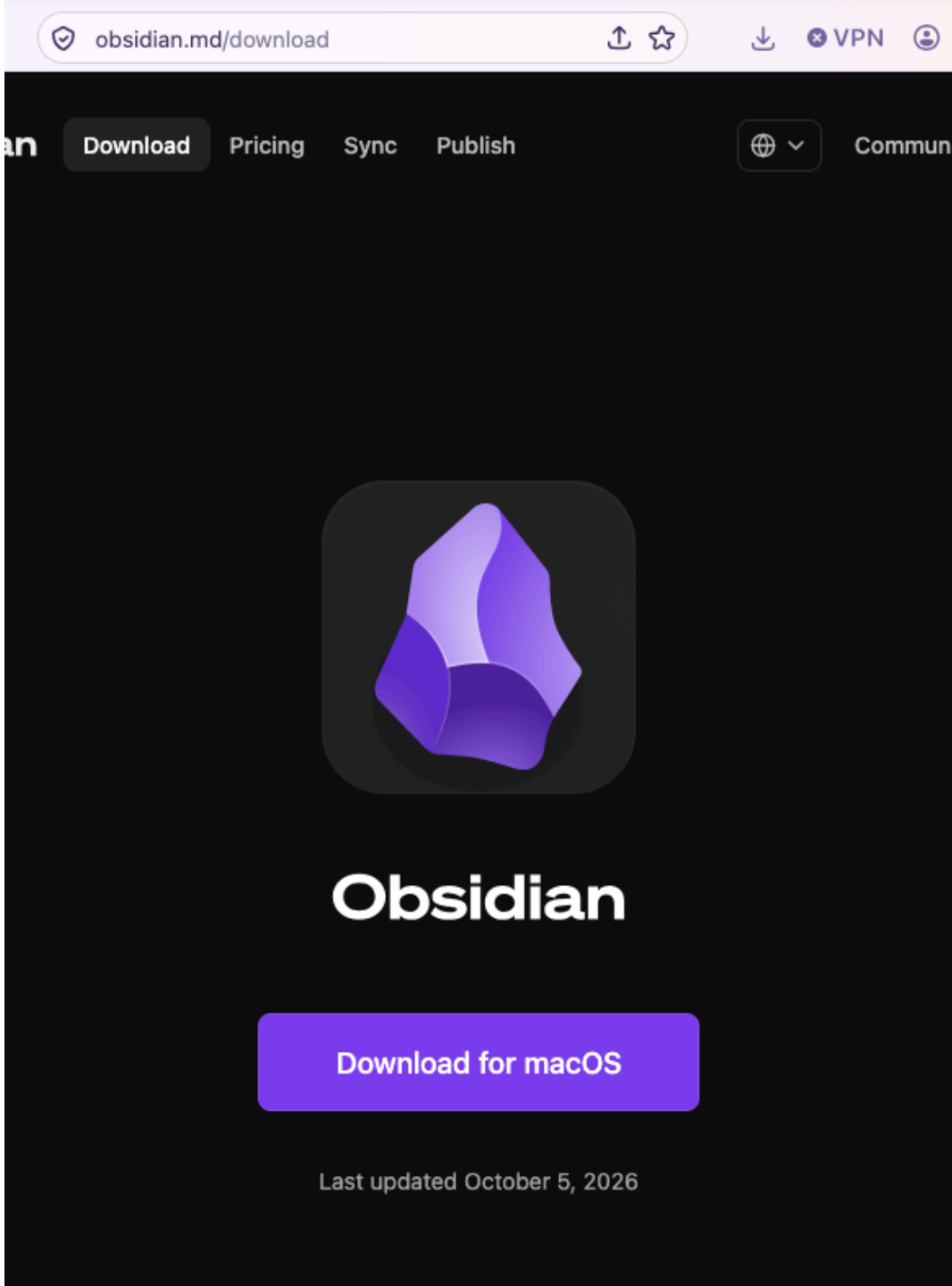
To-do

- [ ] Schedule flights
- [ ] Ask for recommendations
  - [x] Keiko
  - [x] Andrew
  - [ ] Garrett
- [ ] Research ryokans in Kyoto
- [ ] Itinerary

Download Obsidian

https://obsidian.md/download

Available for Windows, Mac, and many Linux distributions.

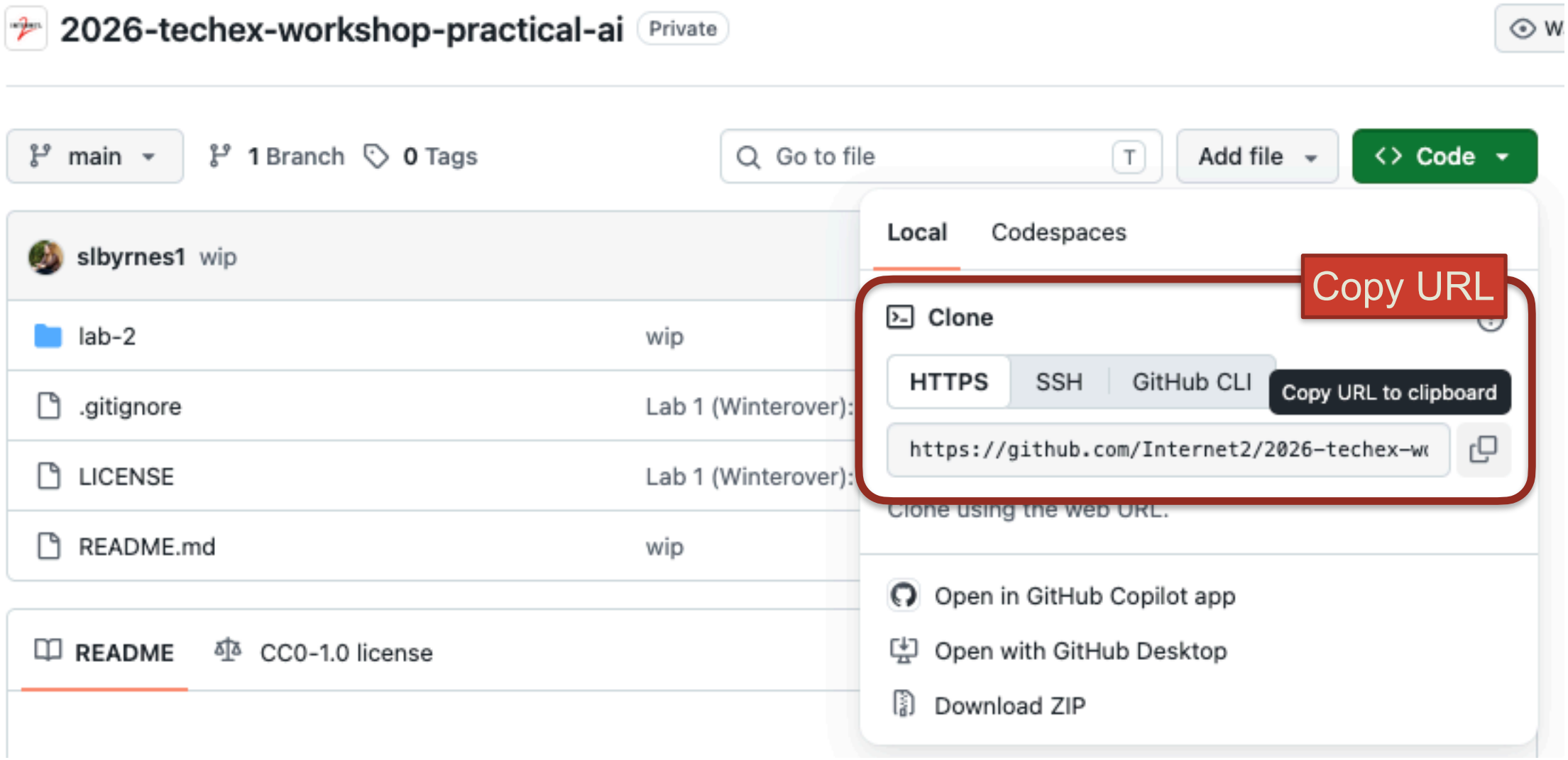


Lab 1 (LLM-Wiki): Building Your Knowledge Base

Ideally, we want to start with some amount of data in a folder on your machine. Two options:

1. Start with the provided fictional data in the “lab-1” folder: https://github.com/Internet2/2026-techex-workshop-practical-ai

2. **Or**, bring your own notes!


2026-techex-workshop-practical-ai
Private
main
1 Branch
0 Tags
Go to file
T
Add file
Code
Local
Codespaces
Copy URL
Clone
HTTPS
SSH
GitHub CLI
Copy URL to clipboard
https://github.com/Internet2/2026-techex-w
Clone using the web URL.
Open in GitHub Copilot app
Open with GitHub Desktop
Download ZIP
slbyrnes1 wip
lab-2 wip
.gitignore Lab 1 (Winterover):
LICENSE Lab 1 (Winterover):
README.md wip
README
CC0-1.0 license

Clone the Repository

1. Clone the repository somewhere on your machine with:

git clone https://github.com/Internet2/2026-techex-workshop-practical-ai.git

2. Then change your directory to it.

```
sbyrnes@Mezzopiano Repos % git clone https://github.com/Internet2/2026-techex-workshop-practical-ai
Cloning into '2026-techex-workshop-practical-ai'...
remote: Enumerating objects: 52, done.
remote: Counting objects: 100% (52/52), done.
remote: Compressing objects: 100% (44/44), done.
remote: Total 52 (delta 7), reused 50 (delta 5), pack-reused 0 (from 0)
Receiving objects: 100% (52/52), 33.25 KiB | 4.16 MiB/s, done.
Resolving deltas: 100% (7/7), done.
sbyrnes@Mezzopiano Repos %
sbyrnes@Mezzopiano Repos % cd 2026-techex-workshop-practical-ai
```

Data for Lab 2

- Optional: Various (fictional) assets contained in a “raw” folder under the “lab-2” folder.
  - Documentation, chat excerpts, excel exports, scripts, etc.

- If you wish to use your own data, you can delete what is in the “raw” folder (or simply create a new folder) and copy in your own assets.

- **When ready, change directory so you are now in the “lab-2” folder (or another folder with your data)**



Quick Note for those using the lab API keys

Prompts, replies, and files you send to the model are not stored on the vLLM host.

Full logging is not enabled, but you should still make sure to not paste in any secrets or anything you wouldn't put in a shared work tool.

What is logged on the host: Timestamps, model, your API key's name (ex. "techex17"), client IP, User-Agent, and token counts.



For Crush,
Select “Nope” to initializing.

Would you like to initialize this project?

~/Repos/2026-techex-workshop-practical-ai/lab-1

When I initialize your codebase I examine the project and put the result into an AGENTS.md file which serves as general context.

You can also initialize anytime via ctrl+p.

Would you like to initialize now?

Yep! Nope

ctrl+c quit • ctrl+c quit • ctrl+g more

sbyrnes@Mezzopiano lab-1 % claude

Accessing workspace:

/Users/sbyrnes/Repos/2026-techex-workshop-practical-ai/lab-1

Quick safety check: Is this a project you created or one you trust? (Like your own code, a well-known open source project, or work from your team). If not, take a moment to review what's in this folder first.

Claude Code'll be able to read, edit, and execute files here.

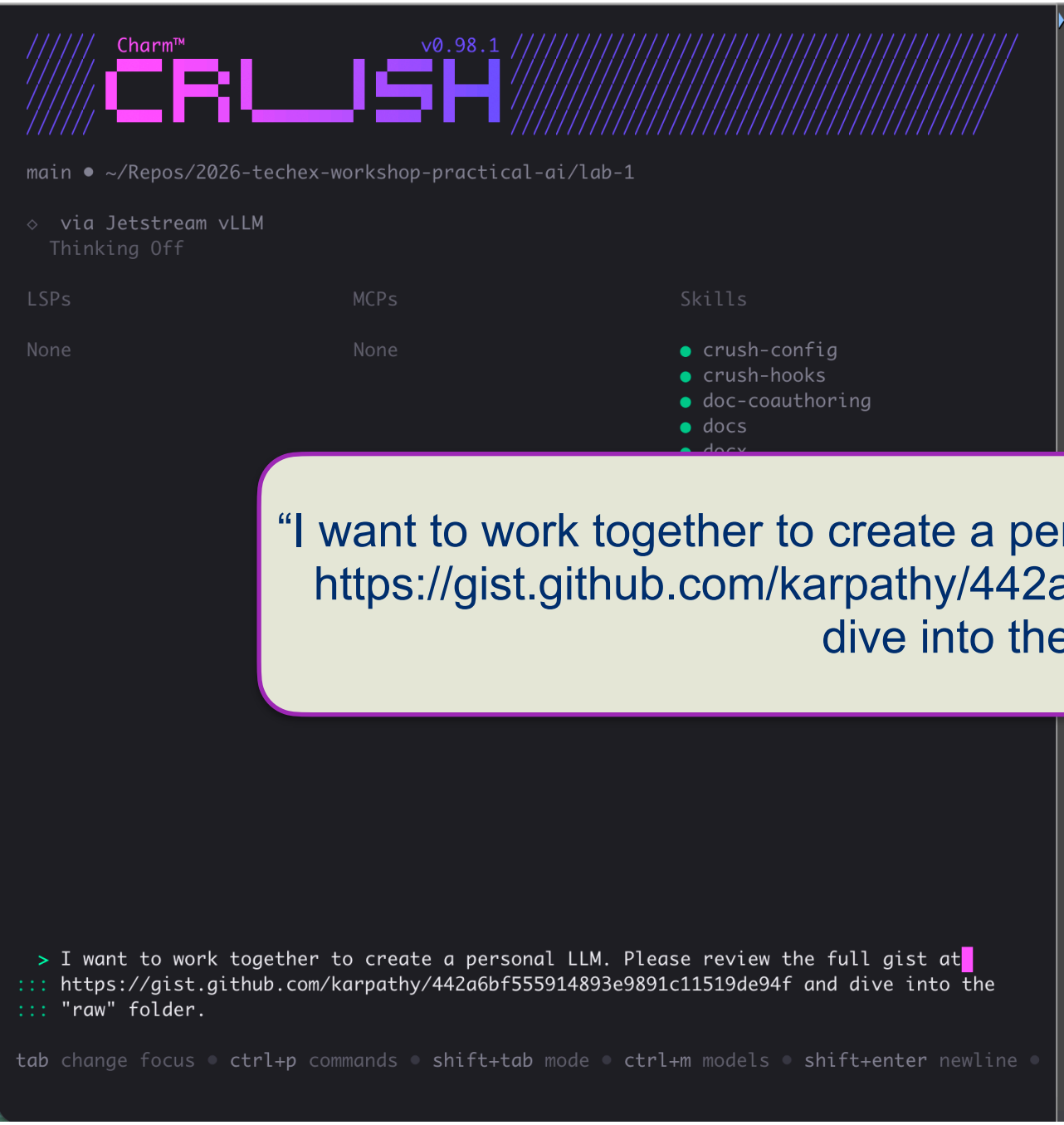
Security guide

❯ No, exit
  Yes, I trust this folder

Enter to confirm · Esc to cancel

For Claude,
Select “Yes, I trust this folder”

Keynote





"I want to work together to create a personal LLM. Please review the full gist at https://gist.github.com/karpathy/442a6bf555914893e9891c11519de94f and dive into the "raw" folder."

I recommend copying the link from the tutorial GitHub page

Charm™ CRUSH ///// ~/Repos/2026-techex-workshop-practical-ai/lab-1 • 17% • ctrl+d open

4	9/14. **Generator runbook lineage.** v1 (250 h, no wind limit) → v2 DRAFT never issued (added the 40 kt rule after the July 2024 blackout) → v3 live (500 h, 20 min pre-warm, 40 kt). All transfers logged in 2026 followed v3; KAT-0512 shows the rule being applied (held at 46 kt).	runbooks/, handover 2024, tickets

Smaller staleness worth pages: SNMP community still `public`, `nightly.py` inventory cross-check stub, `sw-core-1` spare never powered on (a 2021 tradition), `penguin` on the load-shed Science circuit, `bergy` cold-weather jams.

Before I scaffold the wiki (`wiki/` + `index.md` + `log.md` + schema file), a few choices:

● Question How do you want the initial build to go? (+2 more)

Waiting for tool response...

Build style | Schema file | Page conventions | Confirm

? How do you want the initial build to go?

Karpathy personally prefers ingesting one source at a time and staying involved, but the collection is small and you already asked me to dive in.

| **Full first pass now (~15-20 pages)**

Scaffold + one sample ingest first

Strictly one source at a time

> Something else?

[prev tab •] next tab • ↑ up • ↓ do

Lesser findings:
- **Superseded numbers:** UPS runtime is now about 24–25 min (handovers say 30). Generator rotation is 500 h (the archive runbooks say 250). `shaper.sh up` caps uploads at 20 Mbit on an 8 Mbit uplink, so the cap does nothing.
- **Never-done chores:** `penguin` is still on the circuit that gets cut first in a load shed. The spare `sw-core-1` has never been powered on, five years running. `nightly.py` uses SNMP community `public`, and a failed SNMP read shows up as 0% battery.
- **Live documents:** `comms/comms-schedule-pointer.md` says the Handbook, comms Sheet and kickoff slides must be **linked, not copied**. The wiki should honour that: link to them, and only fetch them when answering a question.

Proposed layout

```
lab-1/
  AGENTS.md          # schema: conventions, ingest/query/lint rules, "raw is read-only"
  raw/               # untouched
  wiki/
    index.md  log.md  overview.md
    systems/         # aurora, icebox, penguin, sw-core-1, ups-comms-1, vsat, leo-puck,
printers
    procedures/      # current version of each runbook, with known errors flagged
    people/          # Rosalind, Bjorn, Mei, Luis, Jonah, Hanna, Omar, past IT staff
    issues/          # each finding above, citing its sources, with status
    timeline.md
```

Every claim would cite its raw file (e.g. `raw/chat/comms-it-channel-2026.txt` 2026-02-21). Superseded facts would stay visible with a note on what replaced them, not be deleted.

Two questions before I build it:
- **Who runs the workflow?** I can do the first ingest pass myself, or write `AGENTS.md` so attendees' Crush/Qwen setups can run ingest themselves, with me only checking the result. If this is the lab's model answer, the second seems more useful.
- **Is the layout right?** I'm happy to cut `people/`, or merge the folders into fewer, larger pages.

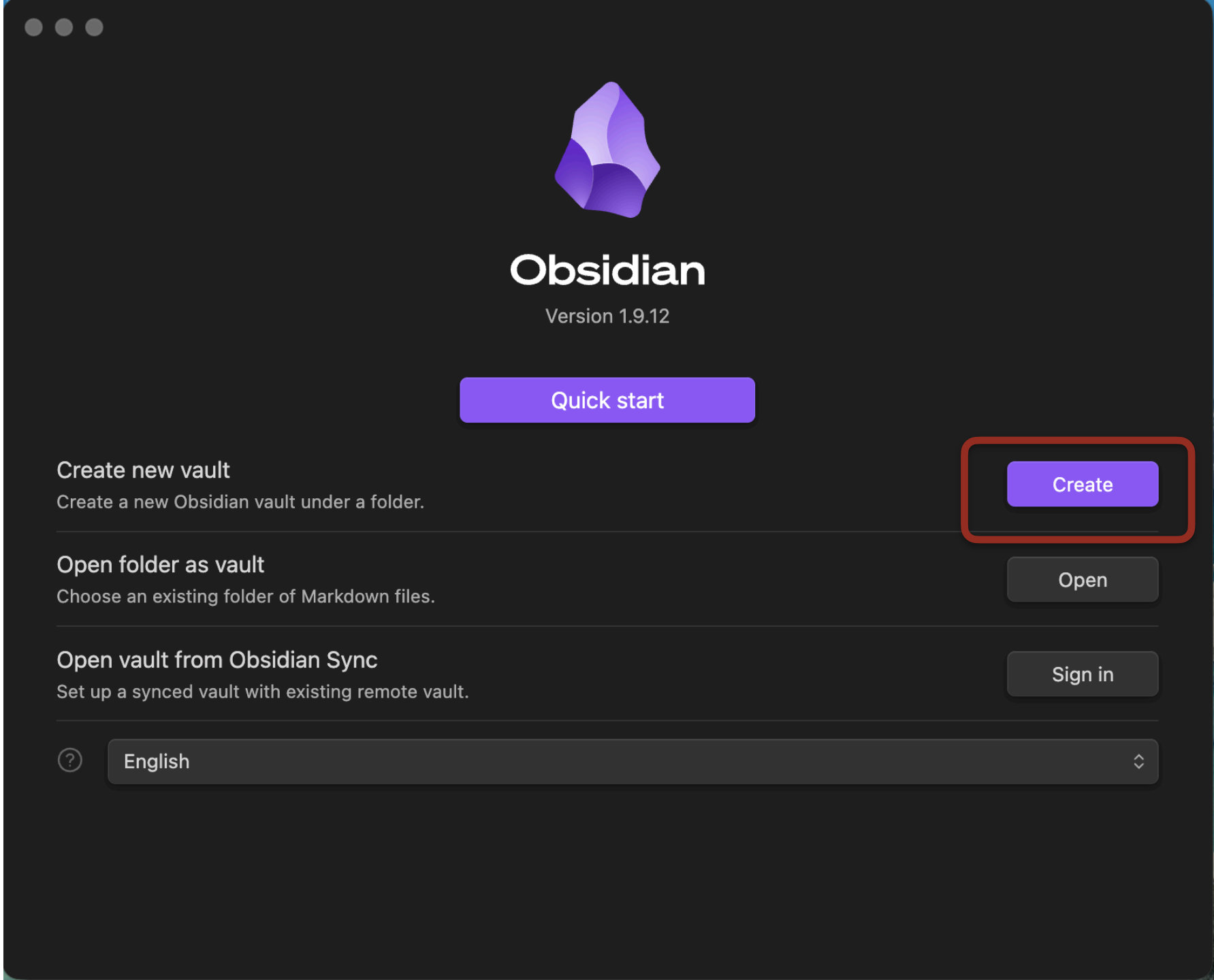
* Cogitated for 1m 18s · done 2:16 PM

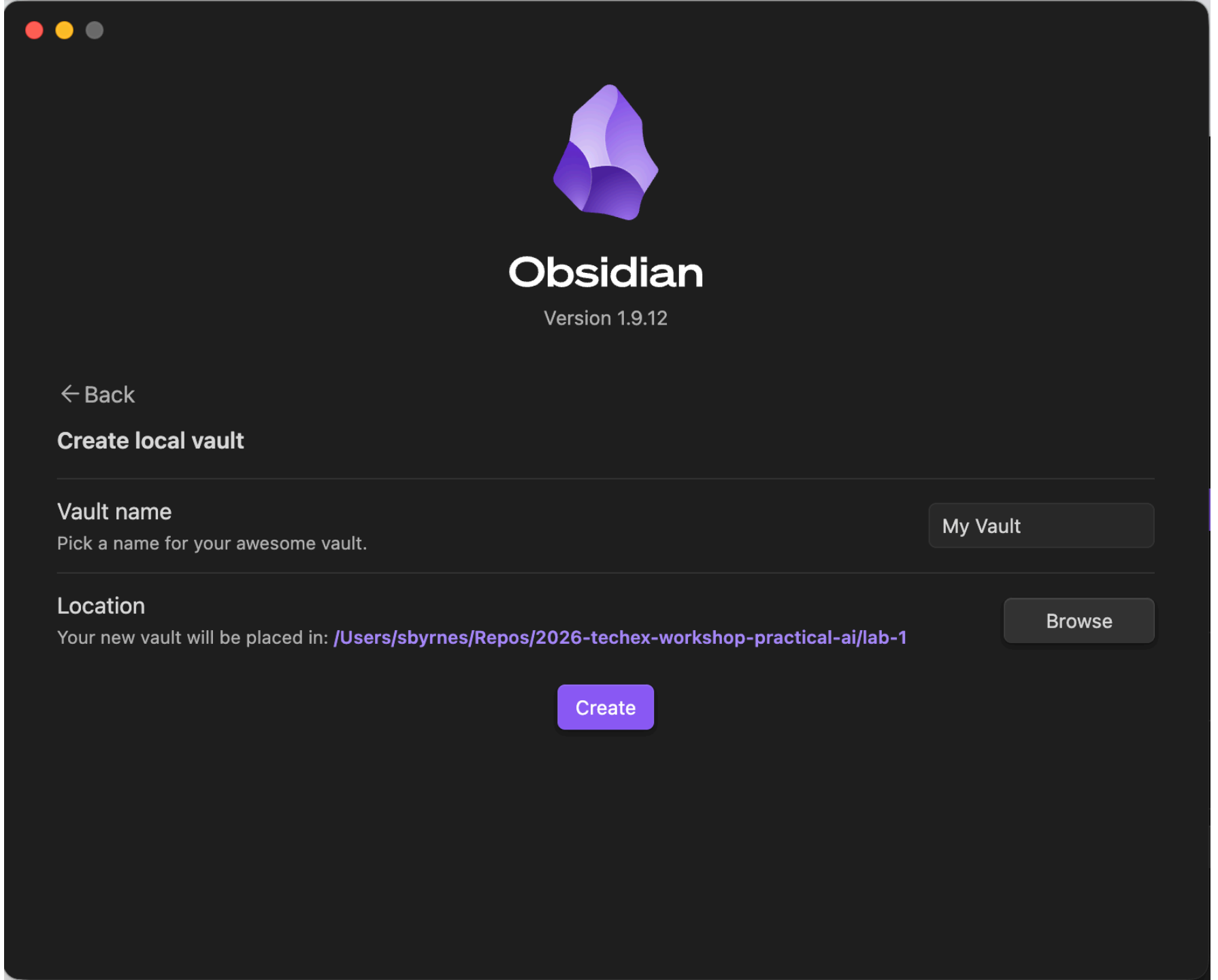
›

▸▸ auto mode on (shift+tab to cycle) · ← for agents

Decide what makes the most sense to you. Feel free to experiment, change directions, and correct the LLM.

Create an Obsidian Vault




Obsidian
Version 1.9.12
Back
Create local vault
Vault name
Pick a name for your awesome vault.
My Vault
Location
Your new vault will be placed in: /Users/sbyrnes/Repos/2026-techex-workshop-practical-ai/lab-1
Browse
Create

AGENTS

My Vault
- Welcome
- raw
  - chat
  - comms
  - handover
  - inventory
  - misc
  - runbooks
  - scripts
  - tickets
  - vendors
- wiki
- AGENTS

lab-1

AGENTS

Katabatic Station IT wiki — schema

This repository is a personal knowledge base for the Katabatic Station IT winterover, built on the "LLM wiki" pattern: the LLM incrementally builds and maintains a persistent wiki between the human and the raw sources.

Layers

- `raw/` — immutable source of truth. Handovers, runbooks, inventory, chat exports, tickets, vendor notices, scripts. **Never modify anything here.** Read and cite; do not edit, rename, or "fix".
- `wiki/` — LLM-owned markdown pages. The LLM creates, updates, cross-links and lints everything here. The human reads; the LLM writes.
- `AGENTS.md` (this file) — the schema. Co-evolved with the human as conventions settle.

Wiki layout

```
wiki/
  index.md          catalog of every page, one line each, by category
  log.md            append-only chronology of ingests/queries/lints
  overview.md       front page: station, systems map, standing rules
  systems/          one page per mach
```

0 backlinks 631 words 4,247 characters

What Might This Look Like?

- Starting an LLM in this folder will now result in an LLM containing your “context” about you (or a fictional you in Antarctica)
- Every conversation and every asset you “feed” it can improve the wiki.
- Send it something important? Tell it to update the wiki or its memory.

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Demo



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Break



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Lab 3: Research Deep Dives



Lab 3: Research & Adversarial Pass

Create a new folder on your machine called “lab-3” and change to it.

Then, load up Crush/Claude.

Lab 3: Research & Adversarial Pass

- We will use the LLM to craft a thorough and specific prompt to pass to a new instance of the LLM.
- Select any research topic you want! Examples are in a slide after this one.
- You can use the following initial prompt to develop a stronger prompt with the LLM.

Lab 3: Example Topics

- Does forced password rotation improve security?
- Do AI coding assistants make engineers faster?
- Has IPv6 adoption stalled?
- Do “learning styles” (visual, auditory, kinetic, etc.) improve learning?
- Is breakfast really the most important meal of the day?
- Does remote work hurt productivity?
- What’s the current state of quantum-safe encryption migration?
- Does willpower run out like a battery?
- How much electricity do AI data centers really use, and who pays for the grid upgrades?
    - Have the LLM evaluate industry-funded vs independent sources.

Using a prompt to create a better prompt

I want to create a prompt that triggers deep research on this topic: ***<topic>***.

I would like the output to be a 2-4 page essay (not including citations). I want all major claims to cite their sources. Sources should be real and verifiable, with peer-reviewed studies prioritized. Low-confidence claims should be flagged. Conflicts of interest and funding sources should be evaluated and considered when deciding to include a study in the output.

Is there anything else we should consider adding to the prompt?

A Possible Reply

Drafted the full prompt in `deep-research-prompt.md`. Key additions beyond your spec and why:

- **Recency + "as of" dates** – figures in this space go stale within a year; stale data masquerades as a solid answer otherwise
- **Measured vs. projected** – IEA, LBNL, and McKinsey numbers diverge a lot; forcing the model to present ranges and explain divergence is the single biggest quality lever
- **"All data centers" vs. "AI workloads"** – most data center power isn't AI; this is the most common conflation in coverage of this topic
- **"Who pays" decomposed by mechanism** – without named mechanisms (network riders, large-load assessments, PUC decisions) the answer tends to be vague like "ratepayers"
- **Peer-review hierarchy with an escape hatch** – the best data (LBNL, IEA) lives in institutional reports, not journals; a hard peer-review-only rule would silently exclude the best sources
- **Anti-hallucination clause** – "real and verifiable" needs to be operationalized: full citation metadata, stable URL/DOI, and an explicit instruction to drop or mark unverified claims
- **COI as weighting, not just disclosure** – say what to *do* with the bias info (weigh against independent sources, present consensus on conflict)
- **Confidence levels with a reason required** – a bare "low confidence" tag is easy for a model to sprinkle; requiring a one-sentence reason forces real evaluation
- **Audience + format** – "2–4 pages" means different things to different models; word count, Markdown, and reader level pin it down
- **Permission to answer with a range** – prevents the model from faking false precision

One thing worth deciding before you run it: which research system this prompt feeds (Claude/ChatGPT deep research, Gemini Deep Research, a custom agent) – some platforms want the citation or confidence-format rules adapted to their output conventions.

Charm™ CRUSH /// ~/Repos/2026-techex-workshop-practical-ai/lab-2 ●…

Hi! Please ingest the research prompt in this folder and begin work.

✓ List .

- /Users/sbyrnes/Repos/2026-techex-workshop-practical-ai/lab…
  - deep-dive-research-prompt.md

✓ View ~/Repos/2026-techex-workshop-practical-ai/lab-2/deep-div…

```
  1  # Deep-Dive Research Task
  2
  3  ## Topic
  4
  5  **What impact does daily LLM usage have on critical think…
  6
  7  ## Goal
  8
  9  Produce a rigorous, well-sourced **two-page essay** (roug…
 10
… (47 lines hidden) [click or space to expand]
```

∴ To-Do 0/7 Researching empirical studies ctrl+t open

> Prrrrrrrr...

esc cancel ● tab focus chat ● shift+tab mode ● / or ctrl+p commands …

Claude Code v2.1.301
Opus 5.5 · Claude Enterprise
~/Repos/2026-techex-workshop-practical-ai/lab-2

❯ Hi! Please ingest the research prompt in this folder and begin work.

Read 1 file, searched memories (ctrl+o to expand)

● I'll fan out research across three independent angles, then draft, then run a fresh adversarial reviewer as the prompt requires.

● 2 background agents launched (↓ to manage)
  ├ Research: offloading & deskilling evidence
  └ Research: counter-evidence & benefits

● I've launched three research agents in parallel:
- evidence of harm (offloading and deskilling)
- counter-evidence and benefits
- differences by user profile, plus real-world signals

When their results are in, I'll write the draft. Then a separate reviewer that had no part in drafting will check the citations and attack the arguments. After I revise, I'll save the essay with that review as an appendix in lab-2/.

* Waiting for 3 background agents to finish

Fetch · from the general-purpose agent 2 of 13
Claude wants to fetch content from dl.acm.org

url: https://dl.acm.org/doi/10.1145/3706598.3713778
prompt: Full title, all authors, venue, date, DOI, sample size, design, key quantitative findings exactly as stated in abstract.

The failed searches don’t necessarily mean there’s a problem. LLM traffic is often blocked or throttled by some websites.

```
            growth? Give exact report title, IEA as publisher, date, and URL. Also note
            any IEA data on US vs global data center electricity.
   ├──── ✓ Search IEA "Energy and AI" report 2025 data center electricity consum…
   ├──── ✓ Search IEA Energy and AI report April 2025 data centers 2030 projecti…
   ├──── ✓ Fetch https://www.iea.org/reports/energy-and-ai
   ├──── × Search IEA Energy and AI report 2024 "300 TWh" OR "AI data centres" e…
   ├──── ✓ Fetch https://www.iea.org/reports/energy-and-ai/energy-demand-from-ai
   ├──── ✓ Fetch https://www.iea.org/reports/energy-and-ai/executive-summary
   └──── ✓ Fetch https://www.iea.org/reports/energy-and-ai/energy-supply-for-ai

6)£^D)(=)F$1_5#

● Agentic Fetch

   Prompt Find EIA (US Energy Information Administration) statements on data center
          electricity use: (1) the 2024 statement that US data centers used about 4%
          of US electricity in 2023 and how many TWh; (2) the 2025 update where EIA
          revised its forecast to data centers using 6.7% to 12% of US generation by
          2028; (3) any EIA statement about AI's share of data center electricity.
          Give the exact article titles, dates, and URLs from eia.gov.
   ├──── × Search EIA data centers 4 percent US electricity 2023 TWh
   ├──── × Search EIA data center electricity 2028 6.7% 12% forecast
   ├──── ✓ Search EIA AI data center electricity share statement
   ├──── × Search site:eia.gov data centers 176 billion kilowatthours 2023
   ├──── × Search EIA "data centers" "6.7%" 2028 electricity generation
   ├──── ✓ Fetch https://www.eia.gov/todayinenergy/detail.php?id=67704
   ├──── × Search EIA data centers 4 percent electricity 2023 176 billion kilowa…
   ├──── × Search EIA "data centers" "4%" electricity 2023 176 billion kilowatth…
   ├──── × Search EIA STEO July 2025 data centers 6.7 percent 12 percent 2028
   ├──── ✓ Fetch https://www.eia.gov/todayinenergy/detail.php?id=63265
   ├──── × Fetch https://www.eia.gov/outlooks/aeo/report/electricity.php
   ├──── × Search EIA "data centers" "4%" of U.S. electricity in 2023 176 billio…
   ├──── ✓ Fetch https://www.eia.gov/outlooks/steo/report/electricity.php
   ├──── ● Fetch 8B2b&F1+=d58€^D
   └──── ● Fetch .+)@...).1f.ff8

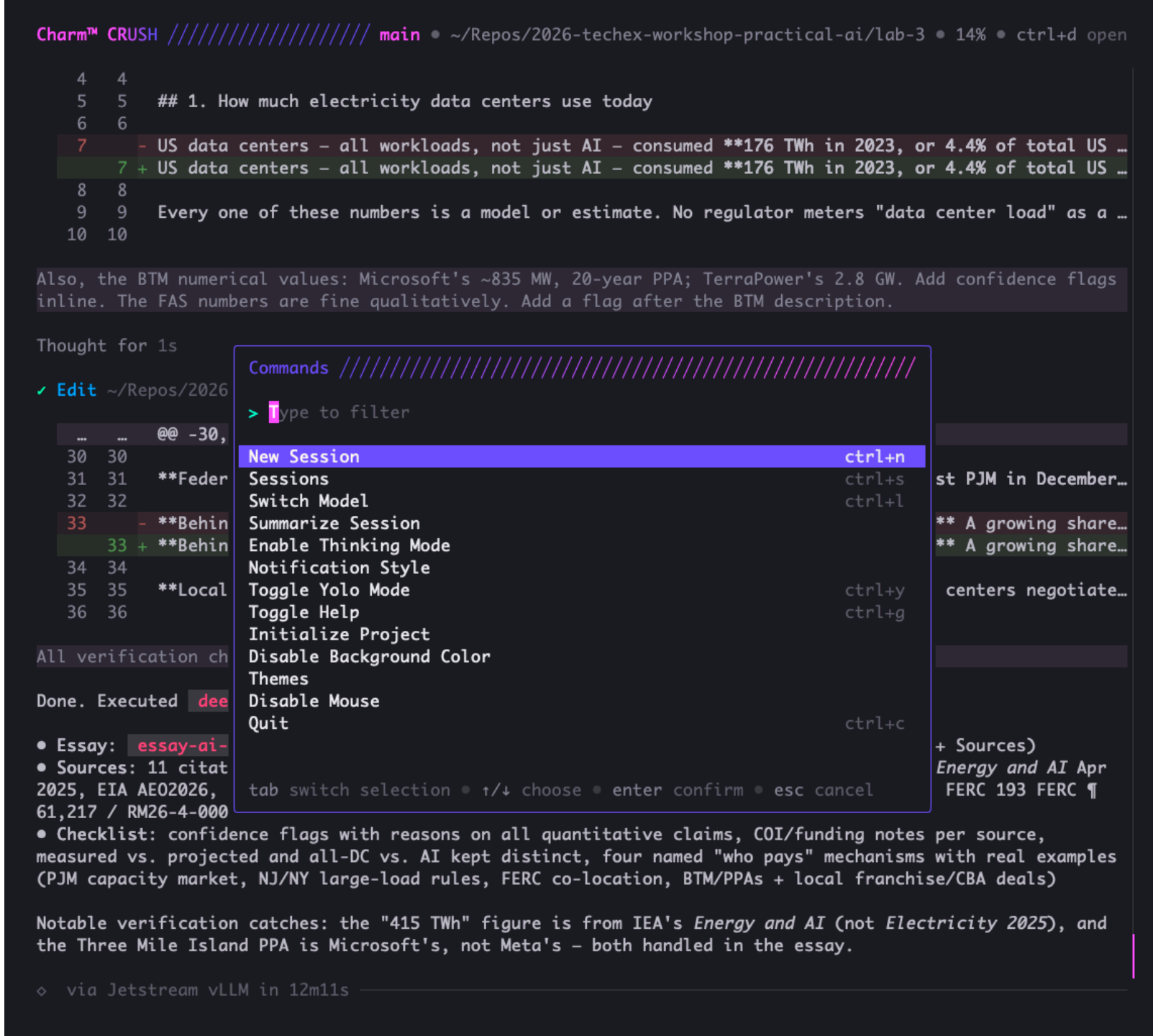
9cEDE7D_&&9)c)D
```

Evaluate; How Did it Do?

- Does the AI go along with the popular answer or push back?
- Did the AI account for who funded a given study? Conflicts of Interest?
- Was the AI nuanced when it came to an “it-depends” topic?

Now let’s try prompting an adversarial pass agent.

Hit / then “New Session” in Crush, or use /clear in Claude



```
  > Please read the "essay-ai-datacenter-electricity.md" essay in this folder and do an
::: adversarial pass. Independently verify all claims for accuracy.
:::
```

```
    ✓ Fetch https://www.nature.com/articles/d41586-025-01113-z
    ✓ Search IEA Energy and AI "United States" "data centres" "4.4%" OR "4.5…

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  To-Do 0/10   ctrl+t close

  Verifying LBNL figures
  Verify IEA Energy & AI figures (415 TWh 2024, 945 TWh 2030, US 183 TWh, GPU share)
  Verify EIA claim (servers ~7% of US commercial electricity 2025, AEO2026)
  Verify PJM capacity auction figures (Monitoring Analytics: $16.4B, $6.5B/40%, $6.2B unbui…
  Verify NJ Data Center Fair Share Act P.L. 2026 c.32 details
  Verify NY EO 62 moratorium and PSC Case 26-E-0045
  Verify FERC dockets EL25-49 and RM26-4-000
  Verify Microsoft/TMI PPA and Meta/TerraPower 2.8 GW deal
  Verify FAS Before Breaking Ground (Cedar Rapids 75%, Lancaster $20M)
  Compile adversarial findings report

> Brrrrr...
```

✓ To-Do 10/10 · completed all

- ✓ Verify LBNL 2024/2025 energy usage figures (176/192 TWh, 2030 projections, AI 55%)
- ✓ Verify IEA Energy & AI figures (415 TWh 2024, 945 TWh 2030, US 183 TWh, GPU share)
- ✓ Verify EIA claim (servers ~7% of US commercial electricity 2025, AEO2026)
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- ✓ Verify Microsoft/TMI PPA and Meta/TerraPower 2.8 GW deal
- ✓ Verify FAS Before Breaking Ground (Cedar Rapids 75%, Lancaster $20M)
- ✓ Compile adversarial findings report

Alright, I'll convey a concise final summary in the chat.

Done. Full report: `adversarial-verification-report.md`

Verdict: Every major number and citation checks out against primary sources (LBNL/OSTI, IEA, EIA, Monitoring Analytics, FERC, NJ/NY records, Constellation, TerraPower, FAS, Lancaster CBA). No fabricated data.

Issues found (7):

1. **Citation error:** FAS "Before Breaking Ground" is cited 2025 but was released Sept. 10, 2026 (it analyzes Nov.–Dec. 2025 agreements, so it can't predate them)
2. **Misleading:** NJ threshold "no greater than 50 MW" – the statute defines large data centers at ≥100 MW peak load; 50 MW is only a cap on the BPU's minimum designation, and the *utilities* (not the BPU) file the tariffs
3. **FERC chronology compressed:** the Dec. 18, 2025 order (193 FERC ¶ 61,217) is the *deciding* order in a show-cause proceeding opened Feb. 20, 2025, not the SCO itself
4. **Off by a day:** RM26-4-000 opened Oct. 28, 2025 (DOE directive Oct. 23), not Oct. 27
5. **Framing:** Cedar Rapids' 75% franchise-fee rebate runs *alongside* the tax rebate, not "in exchange for" it
6. **Likely slip:** Monitoring Analytics SOM "Vol. 1, Mar. 4, 2026" doesn't match evidence (~Mar. 12–13, Vol. 2); the Jan. 5, 2026 auction analysis it leans on is verified
7. **Minor:** NY EO 62 pauses *NYSDEC discretionary* permits specifically, not "state permitting" broadly

The report ends with a 7-item correction checklist for the essay.

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Things LLMs are Bad For (And other Asides)



Things LLMs Are Bad At

- Work you can't easily validate.
  - Even experts miss errors; validating is fatiguing
- Anything deterministic.
- Very large datasets without loss.
  - The context window is finite, and recall degrades in the middle of it
- Being a domain expert.
  - Experts using LLMs find the errors when it comes to their own field.
- Catching its own mistakes. Review by humans and separate model instances helps.
- Long unsupervised autonomy
  - True sandboxing is still unsolved. It also turns off “human in the loop”.

Open Questions About AI

- Model collapse
  - What happens when the training data is increasingly model output?
- Token prices
  - Falling per token, but who pays, and for how long? Pricing is opaque
- Local vs. frontier trajectory
  - Does the gap keep closing? What does a 2027 laptop run?
- Training data, copyright, consent
  - Where did the training data come from?
- Labor and deskilling
  - Junior roles, and our own skills

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Taking It Home



Rung	What	Cost	Institutional data OK?
Laptop, 16 GB x86	3–8B models; speed ⚠ measure on our own machine	$0 + power	Your rules
Laptop, 24–48 GB Apple silicon	27B dense at 4-bit fits (~14–16 GB)	$0 + power	Your rules
Free hosted tiers	Google AI Studio (trains on free tier) · Groq (no retention) · OpenRouter free (may publish prompts; Stripe acquisition pending)	$0 — paid in data, differently each	Usually no
R&E GPUs	Jetstream2 (ACCESS), NRP	Allocation	Public / low-risk only (AUP)
Today's room	2× H100 · vLLM + LiteLLM + a client	~$25–40K per GPU to buy, ~$2–4/GPU-hr to rent, plus ~1–1.5 kW	Public / low-risk only
Hosted open weights	OpenRouter, Bedrock catalog	Per token	Per your agreement
Frontier — consumer	Claude, GPT, Gemini apps	Free / per seat	Usually no; trains by default
Frontier — enterprise / NET+	Same models, contractual terms	Per seat or token	Per your agreement